Common continuous distributions 2

▽ Def. The gamma function is

$$\Gamma(\alpha) = \int_0^\infty e^{-x} \cdot x^{\alpha-1}\, dx, \quad \alpha > 0.$$

▽ $\Gamma(\alpha) = \int_0^\infty e^{-x} \cdot x^{\alpha-1}\, dx$

$u = x^{\alpha-1}$ $\quad v' = e^{-x}$

$\int u \cdot v' = u \cdot v - \int u' v$

$u' = (\alpha-1)x^{\alpha-2}$ $\quad v = -e^{-x}$

$$= \cancel{x^{\alpha-1} \cdot e^{-x}}^{\to 0} \cdot (-1) \Big|_0^\infty + (\alpha-1)\int_0^\infty x^{\alpha-2} e^{-x}\, dx$$

$$= (\alpha-1) \cdot \Gamma(\alpha-1)$$

$$\therefore \Gamma(\alpha) = (\alpha-1) \cdot \Gamma(\alpha-1)$$

- If $\alpha = 1$, then

$$\Gamma(1) = \int_0^\infty \cancelto{1}{x^{1-1}} e^{-x}\,dx$$

$$= -e^{-x}\Big|_0^\infty = 1$$

- If $\alpha = 2$, then

$$\Gamma(2) = 1 \cdot \Gamma(1) = 1$$

- If $\alpha = 3$, then

$$\Gamma(3) = 2 \cdot \Gamma(2) = 2 \cdot 1 = 2!$$

- If $\alpha = n$ is a positive integer, then

$$\Gamma(n) = (n-1)!$$

▽ Def. A RV X is distributed gamma if

$$f_X(x) = \frac{x^{\alpha-1} \cdot \beta^{\alpha}}{\Gamma(\alpha)} \cdot e^{-\beta x}, \quad x > 0.$$

▽ Notation : $X \sim Ga(\alpha, \beta)$, $\alpha > 0$, $\beta > 0$.

▽ $$M_X(t) = \int_0^\infty e^{xt} \cdot \frac{x^{\alpha-1} \cdot \beta^{\alpha}}{\Gamma(\alpha)} e^{-\beta x} \, dx$$

$\beta^* > 0 \iff t < \beta$

$$= \int_0^\infty \frac{x^{\alpha-1} \cdot \beta^{\alpha}}{\Gamma(\alpha)} e^{-\boxed{(\beta - t)}x} \, dx$$

$$= \frac{\beta^{\alpha}}{(\beta^*)^{\alpha}} \int_0^\infty \overset{1}{\underbrace{\cancel{\frac{x^{\alpha-1} (\beta^*)^{\alpha}}{\Gamma(\alpha)} e^{-\beta^* x}}}_{\text{PDF of } Ga(\alpha, \beta^*)}} \, dx$$

$$= \left(\frac{\beta}{\beta - t} \right)^{\alpha}, \quad t < \beta.$$

$$E(X) = \frac{d}{dt} M_X(t) \Big|_{t=0}$$

$$= \beta^{\alpha} \cdot -\alpha \cdot (\beta - t)^{-\alpha-1} \cdot (-1) \Big|_{t=0}$$

$$= \alpha \cdot \beta^{-1}$$

$$E(X^2) = \frac{d^2}{dt^2} M_X(t) \Big|_{t=0}$$

$$= \beta^{\alpha} \cdot \alpha \cdot -(\alpha+1) \cdot (\beta - t)^{-\alpha-2} \cdot (-1) \Big|_{t=0}$$

$$= \alpha \cdot (\alpha+1) \cdot \beta^{-2}$$

$$Var(X) = \alpha \cdot (\alpha+1) \cdot \beta^{-2} - (\alpha \cdot \beta^{-1})^2 = \alpha \cdot \beta^{-2}$$

▽ If $\alpha = 1$ and $\beta = \theta$, then

$$Ga(1, \theta) \text{ "=" } Exp(\theta)$$

▽ THM. Consider independent $X_i \sim Ga(\alpha_i, \beta)$, and let $S = X_1 + \cdots + X_n$. Then

$$S \sim Ga\left(\sum_{i=1}^{n} \alpha_i, \beta\right)$$

$$M_S(t) = \prod_{i=1}^{n} M_{X_i}(t)$$

$$= \prod_{i=1}^{n} \left(\frac{\beta}{\beta - t} \right)^{\alpha_i}$$

$$= \underbrace{\left(\frac{\beta}{\beta - t} \right)^{\sum_{i=1}^{n} \alpha_i}}_{}$$

MGF of $Ga(\sum_{i=1}^{n} \alpha_i, \beta)$